

## MP-01 Conveyor Belt Preventive Maintenance

Category: Maintenance Procedures

Synthetic sample document for public portfolio review.

### DOCUMENT CONTROL

Document ID: MP-01-CNV-220

Revision: Rev 1.0

Owner: Facilities Reliability Team

Applies To: North Campus Distribution Operations

Review Cycle: Every 12 months or after equipment redesign

### PURPOSE

This procedure defines the monthly preventive maintenance routine for the fictional CNV-220 takeaway conveyor used in the packing mezzanine. The goal is to catch belt tracking drift, worn rollers, loose guards, and overheating drive components before they interrupt outbound volume.

The procedure is intended for trained maintenance technicians and shift leads. Production operators may assist with observation and restart confirmation, but they do not perform isolated service work unless the task card explicitly allows it.

### REQUIRED TOOLS AND PARTS

- Lockout kit with personal lock, hasp, and danger tag.
- Infrared thermometer, 6 mm hex key set, belt tension gauge, flashlight, clean lint-free cloths.
- Approved replacement hardware for side guards, nose pulley fasteners, and three standard return rollers.
- Work order printout or digital task card showing the exact conveyor zone and last completed service date.

### HAZARDS AND CONTROLS

- Treat the conveyor as energized until lockout is verified at the local disconnect and start button.
- Keep both side guards installed unless a specific inspection step calls for temporary removal and reinstall them before manual jogging.
- Do not place hands between the belt and roller bed while checking tracking; use a marked reference point and observe from the frame edge.
- If product debris contains broken glass, use cut-resistant gloves and dispose of waste in the designated rigid container.

### PREPARATION

- Confirm the service window with the area lead and post a maintenance in progress sign at both feeder points.

- Stop the conveyor from the local HMI, isolate the disconnect, apply personal lockout, and test the start station for zero response.
- Walk the full conveyor length and inspect belt edges, splice condition, side guides, frame welds, and return roller rotation.
- Clean the nose pulley, tail pulley, photo-eye brackets, and debris traps. Note any product residue that could change tracking or sensor performance.
- Measure motor surface temperature and gearbox housing temperature before adjustment. Compare readings to the baseline values in the task card.
- Check belt tracking by hand-jogging only after all guards are temporarily reinstalled and the area is clear. If the belt drifts more than 5 mm at either edge, adjust the tail tracking bolts in quarter-turn increments.
- Verify roller condition by spinning each return roller. Replace any roller that binds, chatters, or has visible flat spots.
- Re-tension the belt only if the gauge reading falls outside the approved operating band of 32 to 36 units.
- Restore the system, remove lockout, run the conveyor empty for three minutes, then run ten sample cartons through the full path.

## PROCEDURE

- Belt tracks within 5 mm of centerline at empty and loaded conditions.
- No exposed hardware, missing guards, or loose brackets remain after service.
- Motor and gearbox temperatures are stable and do not exceed the prior baseline by more than 8 C.
- Ten sample cartons pass without skewing, snagging, or photo-eye interruption faults.

## ACCEPTANCE CRITERIA

- Record as-found and as-left belt tension readings in the work order.
- List every adjusted component, replaced roller, and reused spare part by quantity.
- If a defect is deferred, create a follow-up corrective work order and note the risk to throughput.
- Obtain shift lead sign-off after the loaded restart test is complete.

## RECORDS AND CLOSEOUT

- A common failure pattern on this fictional conveyor is gradual drift toward the operator side after two weeks of heavy carton volume. In past simulations, the root cause was usually tail roller contamination rather than major frame damage.
- Technicians should avoid overcorrecting tracking with large bolt turns. Small adjustments followed by an empty run and a loaded run give more repeatable results and make later troubleshooting easier.
- When reviewing the service log, compare temperature data by season. Ambient heat on the mezzanine can raise baseline readings in summer without indicating a drive failure.

## EXAMPLES AND NOTES

## APPENDIX 1: REFERENCE NOTES

This appendix page extends the fictional source material with supporting examples, logging expectations, and exception notes so later retrieval tests can traverse a longer document without relying on private or derived content.

Examples on appendix pages are intentionally consistent with the body of the document. They repeat the same operating vocabulary but add different details such as timings, thresholds, role ownership, and escalation triggers.

- Record location code, equipment identifier, shift, and operator initials on each manual intervention.
- If a procedure references a tolerance band, use the tolerance from the most recent approved revision rather than memory or verbal guidance.
- If a required part is unavailable, tag the task as deferred and note the risk to continued operation in the maintenance or service log.
- For training simulations, supervisors may role-play escalation decisions, but the written record should still reflect the documented path in this guide.

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